Momentum Staging

Curriculum Proposal & Repository Framework

I. Repository Architecture

Structuring the future of Data Science research.

Three Pillars of Staging

Version Isolation

Ensures that in-progress curriculum drafts do not interfere with the student-facing main repository.

Peer Review

A collaborative space for Alex and the team to audit materials and propose technical refinements.

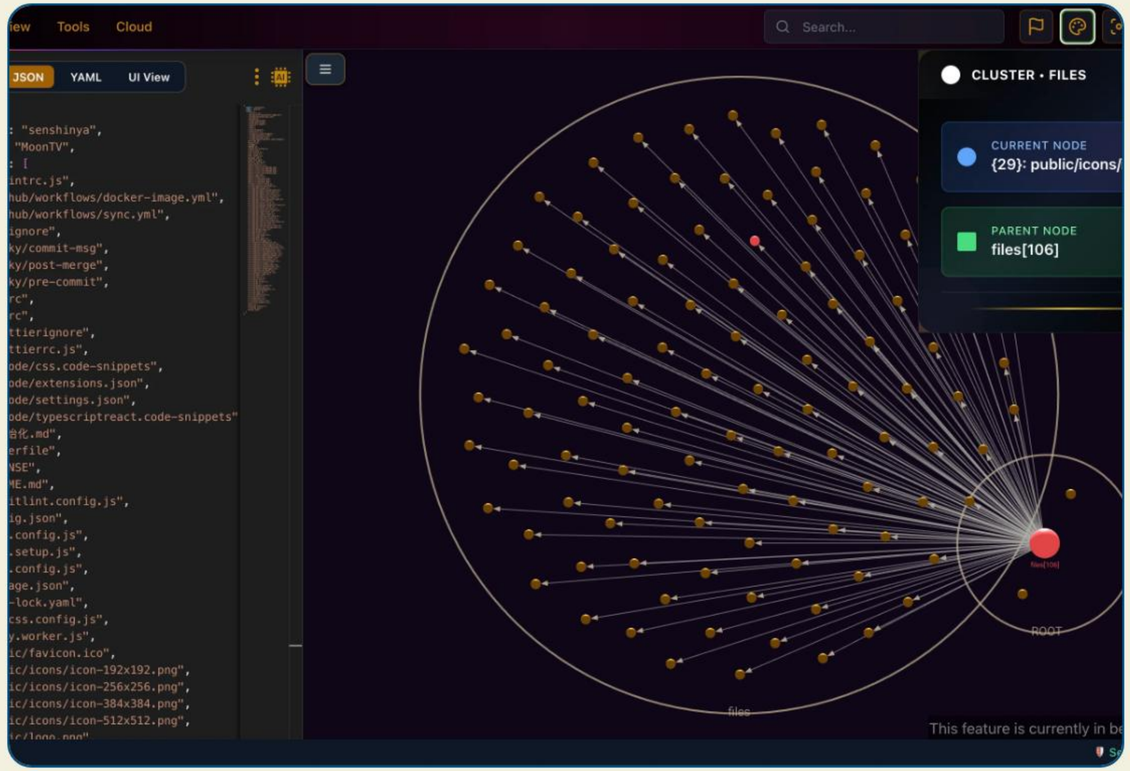
Historical Backup

Centralized storage for all legacy materials and unused datasets for future research reference.

Visualizing Repository Structure

A modular file hierarchy is critical for maintaining over 12 weeks of complex educational data.

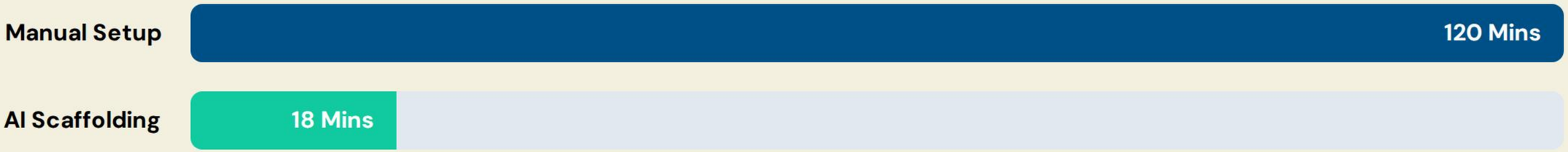
- ✓ **Weekly Materials:** Production-ready folders.
- ✓ **Proposals:** The innovation incubator.
- ✓ **Resources:** Global assets and links.



The Proposal Template Checklist

Section	Required Content	Purpose
Problem Statement	Educational gap analysis	Defines the "Why"
Learning Outcomes	3 specific measurable goals	Defines the "What"
Content Outline	Phase-by-phase breakdown	Defines the "How"
Assessment	Quizzes or Lab assignments	Verifies achievement

Efficiency Gain Analysis



Leveraging GitHub Copilot reduces the initial repository scaffolding time by approximately 85%.

The Review and Merge Flow

Scalability and Traceability

100%

Traceable History

Reliable Data Management

Every change in the curriculum is documented through a standardized template. This ensures that even if team members rotate, the institutional knowledge remains intact and accessible.

Collaborative Innovation

The `proposals` folder is where the next generation of Momentum Research begins. By empowering every researcher to propose new modules, we foster a culture of continuous technical excellence.

Team members can debate the merits of specific algorithms or libraries in the Pull Request comments before final approval.

Data science team collaborating on a strategy board

Template Integration Best Practices

Kebab-Case

Always use lowercase with hyphens for folder and file names to ensure cross-platform compatibility.

Metadata First

Never skip the `Metadata` section in `template.md` as it is the key to repository indexing.

Admin Priority

Only designated leads (e.g., Alex) should merge from `proposals` to production folders.

Vision for the Team

"Structure is the bridge between raw data and actionable knowledge."

— Momentum Research Team Manifesto

Questions?

Let's build the future of curriculum management together.

github.com/Momentum-Research-Team

Image Sources

https://jsonviewer.tools/blog/3d-repo-visualizer.webp

Source: jsonviewer.tools

Thumbnail for www.institutedata.com

https://www.institutedata.com/wp-content/uploads/2023/08/Role-of-a-data-science-leader.png

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